

ALEXANDER BESCH

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[Portfolio](https://alexanderbesch.github.io) (https://alexanderbesch.github.io)



EDUCATION

University of Minnesota, Minnesota Robotics Institute

September 2023 - March 2025

Master of Science, Robotics

Cumulative GPA of 3.81

University of Minnesota, College of Science and Engineering

September 2020 - June 2023

Bachelor of Mechanical Engineering, Mechanical Engineer

Cumulative GPA of 3.83, Dean's List, Graduated with Distinction

EXPERIENCE

Delta ModTech

Electrical & Controls Engineer II

March 2025 – Present

- Designed, programmed, integrated and commissioned optimized complex motion control architectures using multi-axis servo coordination, PLCs, vision systems, and advanced HMI interfaces for \$1+ million projects
- Collaborated with cross-functional teams to ensure seamless mechanical, electrical, and software integration
- Built strong relationships with customers, understanding their needs and delivering tailored automation solutions that exceeded expectations
- Contributed to internal AI initiatives as part of an internal AI team, developing cutting-edge RAG and search-based solutions to streamline information retrieval and decision-making
- Supported installations, testing, and service to ensure reliable system performance and long-term customer satisfaction

PAR Systems

Field Service Engineer – Life Sciences

May 2024 – March 2025

- Designed, programmed, tested, commissioned, installed, and maintained a variety of automation equipment with a wide array of features including HMI, robotic arms, SCARA robots, computer vision, servos, pneumatics, and barcode readers
- Created and executed tests to ensure products met rigorous reliability, safety, regulatory, and customer standards
- Constructed positive relationships with clients and ensured high levels of satisfaction
- Developed and delivered customized training programs and documentation for customers, empowering them to effectively operate and troubleshoot automation systems
- Assisted with integration of AI/LLMs into business workflows while maintaining data security

University of Minnesota

Teaching Assistant – Robotics, University of Minnesota, Twin Cities

January 2024 – May 2024

- Created, revised, and graded examinations to accurately reflect students' knowledge of course material
- Conducted lectures to review previous exams and answered questions to promote student success
- Interacted with students to create a classroom environment that fit individual learning needs

ArchKey Solutions

Engineering Intern – ProKey Facilities / ArchKey Technologies

February 2021 – May 2024

- Managed procurement, and installation of mechanical equipment on large scale commercial buildings
- Maintained AutoCAD blueprint database in conjunction with physical site changes
- Assisted the planning, logistics, and project management of various multi-million-dollar construction projects from beginning to end
- Streamlined project management processes using Python and Microsoft VBA
- Supervised first year intern throughout summer program and provided feedback, and coaching

SKILLS

Languages: Python, Rockwell Automation, RSLogix, C#, VisualBasic.NET, C++, MATLAB, JavaScript

Tools/Libraries: OpenCV, PyTorch, TensorFlow, LangChain, W&B, Git/GitHub, Ultralytics

Software: Microsoft Office, Studio5000, Cognex VisionPro/Spreadsheet, AutoCAD, SolidWorks, Visio, Bluebeam, FT ViewStudio

Operating Systems: Windows, Linux, MacOS

Other: Project management, Customer Interaction and Relationship Building, PLC/HMI Development, Ethernet I/P, OpenAI API